

* Loops:

⇒ Loops in python are used to execute a block of code repeatedly until a certain condition is met.

Python mainly provides three types of loops:

① For loop ② While loop

③ nested loops

- For loop inside For loop
- For loop inside while loop
- while loop inside while loop
- while loop inside For loop

① For loop

→ A For loop in python is a control-flow statement that allows you to iterate over a sequence (like a list, tuple or string) or other iterable objects.

- For loop with sequence.
- For loop with range.

Syntax

For i in sequence - name: # i: Iterative variable
statements

- for loop with range
initialization
condition
increment / decrement

range (start value, stop value, step size)

range (i=0, i<u, i=i+1)

name = "Kiran"

Print (name)

Ex: for i in name :

Kiran mode

Print (i)

OLD

	name[i]	for i in enumerate (name):
K	'K'	
i		
r		print(i)
a		print ("Position = {i[0]}, character = {i[1]}")
n		

OLD

- We can print the name with their position and the character we can use the enumerate.

(0, 'K')

pos = 0, char = K

(1, 'i')

pos = 1, char = i

(2, 'r')

pos = 2, char = r

(3, 'a')

pos = 3, char = a

(4, 'n')

pos = 4, char = n

Ex 1: $p = 0$
 for i in name:
 print(f"pos = {p}, value = {i}")
 p = p + 1

O/P

pos = 0, value = k	pos = 3, value = q
pos = 1, value = i	pos = 4, value = n
pos = 2, value = j	

Ex 2: for i in range(1, 5, 1):
 print(f"i = {i}")
 print("Hello", "Kitan")

O/P

i = 1	i = 3
Hello, Kitan	Hello, Kitan
i = 2	i = 4
Hello, Kitan	Hello, Kitan

Ex 3: Print numbers from 1 to 20
 for i in range(1, 21, 1):
 print(i)

O/P

1	Ex 4: print odd numbers from 91 to 100 for i in range(91, 100, 2): print(i)		
2			
3			
4			
5	O/P		
6			
7			
8			
9	91	95	99
10	93	97	

Ex 5: Write a program to check the from 1 to 10 is even or odd
for i in range(1, 11, 1):
 if (i % 2 == 0):
 print(i, "-> Even Number")
 else:
 print(i, "-> Odd Number")

O/P

1 -> Odd number	6 -> Even number
2 -> Even number	7 -> Odd number
3 -> Odd number	8 -> Even number
4 -> Even number	9 -> Odd number
5 -> Odd number	10 -> Even number

Ex 6: Write a program to calculate the sum of numbers, sum of even numbers, sum of odd numbers from 1 to 10.

ts = 0

es = 0

os = 0

for i in range(1, 11, 1):

 ts = ts + i

 if (i % 2 == 0):

 es = es + i

 else:

$$os = os + i$$

print("Total sum = ", ts)

print("Even sum = ", es)

print("Odd sum = ", os)

Qp

Total sum = 55

Total even sum = 30

Total odd sum = 25

Ex 7: Write a program to print tables of given number O/P $3 * 1 = 3$ | $3 * 2 = 6$ | $3 * 10 = 30$?

n = int(input("Enter a number: "))

for i in range(1, 11, 1):

print(f"{n} x {i} = {n*i}")

O/P

Enter a number : 45

$$45 \times 1 = 45$$

$$45 \times 10 = 450$$

$$45 \times 2 = 90$$

$$45 \times 3 = 135$$

$$45 \times 4 = 180$$

$$45 \times 5 = 225$$

$$45 \times 6 = 270$$

$$45 \times 7 = 315$$

$$45 \times 8 = 360$$

$$45 \times 9 = 405$$